Results

Atmospheric pressure on day of experiment: 101.6 kPa

TABLE 1: SATURATION LAB RESULTS

Gauge pressure, ***P*1** (kPa)	Absolute pressure, ***P*1**,abs (kPa)	Temperature, ***T***exp (°C)	Resistance, ***R*** (Ω)
0	101.6	100	138.71
100	201.6	120	146.04
200	301.6	134	151.17
300	401.6	144	155.17
400	501.6	150	157.91
500	601.6	158	160.71
600	701.6	162	162.13

Aim

To study the relationship between the saturation temperature and pressure of water (the point at which water vaporises) and to evaluate the accuracy of saturation data obtained using a laboratory apparatus by comparing measured values to steam-table data.

Equipment

- Armfield saturation-pressure apparatus (boiler with sight-glass & viewing port)
- Orange power/control console with heating element
- Valves: filling-point (with key), isolating valve, drain valve
- Pressure gauge P (gauge pressure P1P_1P1)
- Platinum resistance thermometer T1 (experimental temperature TexpT_{\text{exp}}Texp)
- Barometer to measure atmospheric pressure (e.g., Fortin barometer)
- Purified/de-ionised water

Method

1. **Prepare the rig**: Ensure isolating and drain valves are closed; console power off.
2. **Fill**: Open the filling-point valve with the key; add de-ionised water to mid-height of the sight-glass. Leave the filling-point valve open.
3. **Purge air**: Power on and heat to vigorous boiling so steam escapes from the filling point; maintain a steady steam flow until temperature stabilises—this expels air.
4. **Close system**: Close the filling-point valve to create a closed, constant-volume vessel containing a saturated water/steam mixture.
5. **Manage thermal lag**: Return heater to maximum for ~2 min, then lower power and wait for the **T1** reading to stabilise so the sensor temperature equals the fluid temperature (reducing thermal lag).
6. **Record readings**: Note TexpT_{\text{exp}}Texp (from **T1**) and P1P_1P1 (gauge) once steady.
7. **Repeat**: Increase heat in steps and repeat stabilise-and-record until P1≈700 kPaP_1 \approx 700\ \text{kPa}P1≈700 kPa, obtaining 7–8 data points.
8. **Convert pressure**: Compute absolute pressure for each point using
   Pabs=Patm+P1P_{\text{abs}} = P_{\text{atm}} + P_1Pabs=Patm+P1
   where PatmP_{\text{atm}}Patm is measured with the barometer.

9. **Comparison task (for analysis)**: Plot TTT vs PabsP_{\text{abs}}Pabs for the experiment and overlay the water saturation line from steam tables; assess agreement and discuss any deviations with reference to thermal lag and other experimental factors.
10. **Shutdown**: Turn off heat; slowly open the isolating valve to bleed steam and reduce pressure/temperature; once cooled, leave the isolating valve open.

Analyse your results as follows:
1. Convert the gauge pressures for each equilibrium stage to absolute pressures and plot a graph of temperature (°C) versus absolute pressure (kPa). Use Excel to plot your graphs.

2. On the same axes plot a graph of the saturation temperature/pressure values from the steam tables in the Appendix of your textbook 1 (a copy is provided on Blackboard).

3. The relationship between saturation temperature, (°C) and absolute saturation pressure, (kPa) can be considered to be of the form:

$$T_{\text{sat}} = AP_{\text{sat}}^{n}$$

where A and n are constants. Using the graphs you have plotted above, use the curve fitting capabilities of Excel (or similar) to determine the values of A and n for your experimental data AND for the steam table data. Clearly state these values in your Results.

To determine the errors between the experimental data and the steam table data it is necessary to determine the theoretical saturation temperature Tth for each experimental pressure reading. To do this, use the steam tables and interpolate to determine the theoretical saturation temperature Tth for each recorded experimental pressure point P1, abs. Record your results in a table similar to Table 3.

For each data point, determine the error percentage between your experimental data, Texp, and the interpolated tabledata, Tth, using the equation below.

$$Error\ \% = \frac{T_{\text{exp}} - T_{\text{th}}}{T_{\text{th}}} \times 100$$

From this determine an average error for your results.

Experimental Pressure ***P***1, abs (kPa)	Experimental Temperature ***T***exp (oC)	Theoretical Temperature ***T***th (oC)	Error (%)